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Effective Approach to Reduce Patient Long Queues in Hospital Emergency Department

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***Abstract*—Patient queues in hospital Emergency Departments (EDs) have been a longstanding challenge, leading to compromised patient care quality, increased wait times, and heightened dissatisfaction among both patients and healthcare providers. This abstract presents an effective approach aimed at reducing patient queues in EDs through a multifaceted strategy. This study utilized queuing theory to reduce the long queuing issue using St. Elizabeth hospital in Eastern Cape provinces as a case study. The quantitative methodology is used to obtained information from the respondents in terms of optimizing triage processes using technology-driven assessment tools to swiftly identify and prioritize patients based on the severity of their condition. The results of the findings on shows that utilizing of the conventional queue method in hospitals does not only leads to wasting of patient time but also complicate some patients' medical condition.**

Keywords—patients, OPD, queueing theory, patient long queue, patient flow, resource allocation, hospital operations

I. INTRODUCTION

In today's fast-paced world, the efficiency of healthcare services plays a pivotal role in ensuring patient satisfaction and well-being. Nowhere is this more apparent than in hospital Emergency Departments (EDs), where long queues often characterize the patient experience, leading to increased wait times, overcrowding, and decreased quality of care. Addressing this issue requires a multifaceted approach that encompasses various strategies aimed at streamlining processes, optimizing resource allocation, and enhancing patient flow.

Healthcare systems in Africa have been facing a massive increase in demand, which has led to a continuous need to improve and optimize operational processes and quality control methods. Many hospitals have begun to encounter logistical and operational problems that previously did not present as a noticeable concern [1]. One of the main problems in hospitals is long queues due to low capacity and high rate of patients.

Emergency Department is the first station that patients encounter when entering the hospital before referred to the appropriate medical practitioners [2]. This department is always congested with long queues of different patients with varieties of health conditions which are been attended to in most cases by few medical practitioners. Consequently, caused delay in patient waiting time and even some unattended to [3].

Patient booking and queueing during an emergency is one of the challenging issue many hospital managements in South Africa are facing [4]. Numerous efforts have been invested in addressing these pressing issues but none have reached conclusive solutions. The research conducted by [5] indicated the demand for emergency services has increased steadily over the past decade, leading to overcrowding and prolonged wait times in EDs worldwide. This has made many patients not to be attended to and some infected with other diseases. There is urgency of finding effective solutions to mitigate the challenges associated with patient queues in the ED setting. By implementing evidence-based interventions and leveraging innovative technologies, healthcare providers can significantly reduce wait times, improve patient outcomes, and enhance overall operational efficiency.

Many hospitals are using Hospital Information System (HIS) technology to address challenging issues. In 2013, The Indonesian government ordered utilizing the HIS as a potential mechanism to assist emergency patients in a public medical clinic, and this has shown a significant improvement in reducing the long queue in Indonesian government hospitals. The adoption of this technology is still at infant stage as many developing countries struggles in acquiring this technology.

In South Africa, many government hospitals and private hospitals are still experiencing these long queues during the patient emergency crisis due to resource constraints [6]. This has led many patients to be devastated and some patients left the hospital unattended. Consequently, leads to patient death or compounded health complications [7]. Hence, it is imperative to provide a measure to assist hospital management in providing solutions to address this challenging issue.

This paper explores the key components of an effective approach to reducing patient long queues in hospital emergency departments. Drawing upon insights from research literature including process optimization, capacity management, staff training, and the utilization of advanced technologies By adopting a queuing theory that addresses both systemic issues and patient-centric concerns, healthcare organizations can achieve sustainable improvements in ED performance while prioritizing the delivery of timely and high-quality care to those in need.

Based on the above, the following research questions emerged;

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Research Question (RQ1): How can a true reflection of what patients are facing be known from EDs?

Research Question (RQ2): What can be done to reduce the long queue in hospitals to provide adequate health care delivery for patients?

To address the research questions above, the contributions of this study are as follows:

- Utilization of quantitative research (questionnaire) approach to evaluate causes of long queues at EDs in hospitals.
- Conceptualization of queueing model variables in questionnaire to deduce the issues causing long queues in hospitals.

The remainder of this paper is arranged in the following order: Section II provides related work and the conceptualise framework. Section III presents a detailed public hospital long queue methodology; Section IV discusses the results and findings of the study. The concluding remarks are shared in Section V.

II. Related Works and Theoretical Background

The work done by [8] addressed the challenging issue of the significant delay that arises due to patient congestion in hospitals which is because of inadequate medical clinics and facilities. The authors utilized a qualitative method to address this issue. The qualitative method was used in this examination due to the fact there has been now no longer any general device for reading and measuring overcrowding. From the results obtained, the specialty wards of the studied clinic were experiencing high referrals of patients for various reasons.

Author in [9] believes that many patients complained about the amount of time it takes to get to the hospital and to visit hospital staff, especially doctors as one of the causes of the long queue in the hospital. The authors utilizes Mobile Appointment Scheduling System (MASS) to address this issue with the objective of enhancing appointment scheduling in hospitals by allowing patients to register for appointments through mobile phones at their own time wherever they are and make an appointment in their desired slot of time. Based on the dynamism and complexity of healthcare planning systems using the simulation model, the results show that the combination of changing doctors' working hours and patient appointment times can reduce patient waiting times by up to 73%.

Furthermore, [3] addressed different issues in all medical care units which ranges from appointment systems, progressive patient flow, patient records, and examination procedures that has caused significant increase in waiting time at each level, leaving patients highly dissatisfied and delayed medical care. The Six Sigma method was applied in Hospital Information Systems (HIS) heart disease department. From their analysis, they suggested several improvements to the existing system, including a check-in desk, designation of a caregiver to accompany the patient, additional reception staff training on phone calls, and call handling manners. The studied sample consisted of the managers and key people of the clinic wards, the need for previous coordination and arranging an appointment to conduct interviews with them was one of the study limitations. Participants in this study were the managers of the clinic wards, in some cases, it was possible that they provided unreal positive feedback and responses and did not tell the truth.

A. Conceptualized Framework

Queueing theory: Queueing theory can be used to resolve the issue pertaining to the patient long queue in hospitals in an orderly manner by giving supervisors proficient arrangements and devices for keeping up with and further developing execution [10]. The dynamics of service systems are extremely complex because of this variability and the interaction between the arrival and service processes. As a result, without the assistance of a queueing mode, it is impossible to anticipate levels of congestion, or the amount of capacity required to achieve some desired level of performance. This is supported by research in [11] thorough writing survey concerning the use of the queueing hypothesis in medical care. However, as of late, there has been an expanding center around issues that are well-defined for the ED, alongside a dire need to tackle them. In 2018, [12] looked at the commitments of Queueing Theory (QT) in demonstrating ED and surveying the qualities and constraints of this application: the ED is the main station of an orderly survey of strategic ways to deal with help process that patient experience while entering the clinic framework as shown in Fig.1.

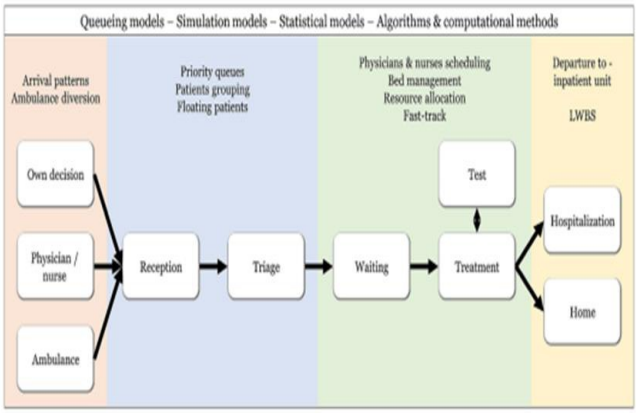


Fig. 1. Patient medical care queueing framework.

Fig.1 presents the patient course through the medical care framework and accentuates the requirement for coordination between interfacing offices. At each period of the cycle, we show the applicable related queueing hypothesis viewpoint. The current paper leads a deliberate writing survey to respond to the essential inquiry of which hypotheses and models can give proficient arrangements and instruments to keep up with and further develop ED execution. The subsequent section gives a detailed explanation of the proposed methodology used in this research.

III. Proposed Hospital Queueing Methodology

The research method used in this research is quantitative method. The questionnaire is set using the variable on the queuing theory and the questionnaire is distributed to patient at St. Elizabeth hospital, Eastern Cape, South Africa. The patient's consent is sought and none of the patients are forced to answer the questionnaires. The response of the participants is analyzed using SPSS software and the result of the findings is discussed.

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A. Data Collection

In this study, questionnaires are used to obtain the information from patients. The questionnaire is in 1-5 Likert scale. The questionnaires are a set of printed or written questions with a choice of answers that can be obtained through written responses. The total sample of participants used in this study is twenty-one (21). The data was collected from patient from 18 years and above, the reason for starting from age 18 years old patient is that in South Africa, the patient is considered as adult and can take decision by themselves and give a valid answer on the questionnaire given.

B. Population and Sample

The hospital that would be used in this study was chosen using the convenience sampling method. A sample of registered hospital patients was chosen using a straightforward random sampling technique. Participants who were available and willing to take part in the study were selected using a simple purposive sampling method. The study included both men and women. To avoid any inconvenience, participants were given paper-based questionnaire at the hospital

IV. Results

A. The Age Distribution of the respondents

The question regarding the respondents' age was asked. Fig. 2 shows the responds results.

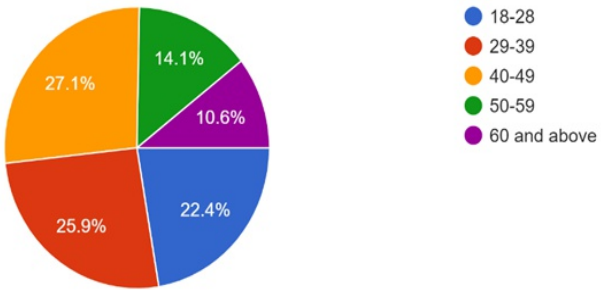


Fig. 2. Age chart presentation

B. Gender of the Respondents

The question regarding the respondents' gender was asked. Below shows the responds results.

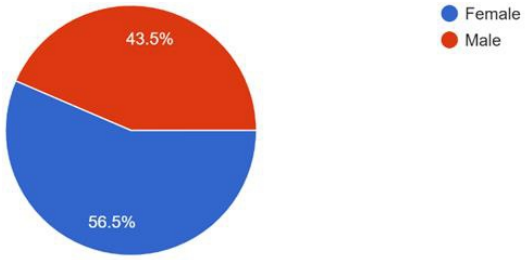


Fig. 3. Respondent Gender.

Fig. 3 shows that 56,5% of the respondents were female and 43,5% were male respondents.

A graphical representation of these results is illustrated on a bar graph in Figure 4.2. These results show that even though gender equality was tried, yet there were more female participants who were more than males.

C. Staff and patients respondents

The question regarding whether the respondent is a hospital staff member or patient. Fig. 4 shows the corresponding response of the respondent.

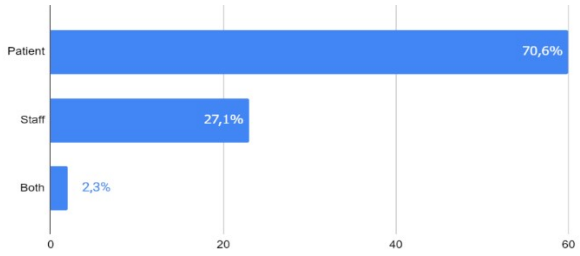


Fig. 4. Staff and patient's graphical presentation.

Fig. 4 shows that 70.6% of respondents were patients and 27.1% were the staff. The percentage 2.3% shows they were both the staff and patients. This question was posed to see if the staff and patients are treated equally in terms of the medication being prescribed fully, and with respect, and to see if they are treated with care.

D. Respondent distance from house of to the hospital

The question was asked the patient to know the distance from their houses to the hospital. Below illustrate their responses.

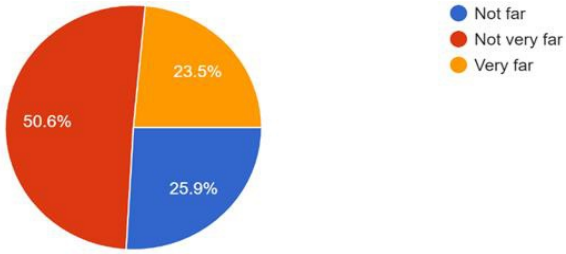


Fig. 5. Respondent responses on distance from their houses to the healthcare center.

Fig. 5 shows the majority (50.6%) do not live very far, 23.5% live very far and 25.9% live at least far. Normally people will wake up very early, as early as 5 0'clock in the morning, to go to the hospital so that they can be assisted first but that is very dangerous for people who live far, and others will have to walk to the hospital.

E. Respondent rating of the hospital service respondents

The question regarding the rating of the hospital services was asked. Fig. 6 illustrates their responses.

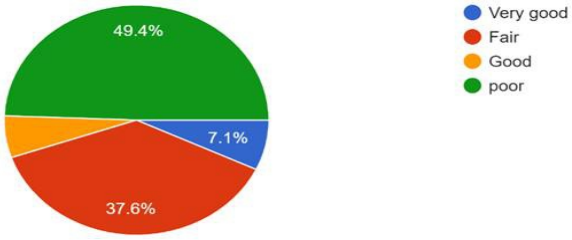


Fig. 6. Respondent response regarding the rating of the hospital services they received.

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Fig. 6 shows that 49,4% of the respondents said it was poor, 37.6% of the participants rate the service as fair and 7.1% as very good and 5.9% shows it is good. It also shows that the highest rates are poor service ratings (49.4%).

SECTION B: PATIENT CARE PROVIDED BY STAFF AT A MEDICAL FACILITY (PATIENTS ONLY)

A. Respondent views on the Effect of the long queues

The question on the effect of long queues on patients was also asked. The responses are shown in Fig. 7.

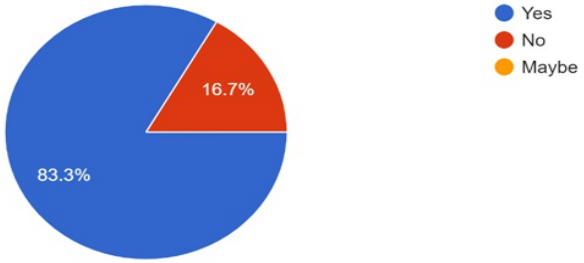


Fig. 7. Respondent patient response on the effect of long queues.

Fig. 7 shows 83,3% of the patients are affected by long queues and 16.7% are not affected at all. With this highest percentage of people being affected, there is the highest possibility of the virus spreading, people getting sicker and people dying in queues.

B. Respondents patient response on joining wrong queue

The question was asked from the patient to know whether they have been on wrong queue before at hospital. The responses are shown in Fig. 8.

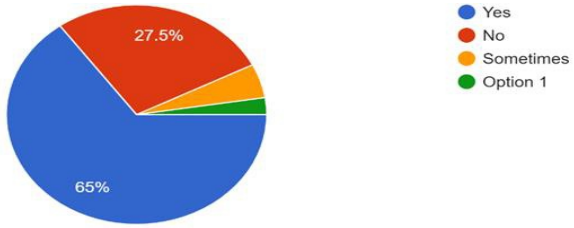


Fig. 8. Respondent patient response on wrong queues

Fig. 8 shows that 65% stand in wrong queues and later figures that they are in wrong queues, which is very devastating and 27.5% they have not queued on wrong and the other 5,5% it happens sometimes to some of them. The patient can stand in line for registration with the ticket. Despite the simplicity of this procedure, many patients nevertheless forgo pre-check and proceed directly to the registration counter without obtaining a pre-check ticket. Patients will not be able to register in this circumstance, and they will have to return to redo the pre-check.

C. Directions to join queues respondents

Patients were asked whether to know if they got directions as to where to join the right queue. The corresponding response is shown in Fig. 9.

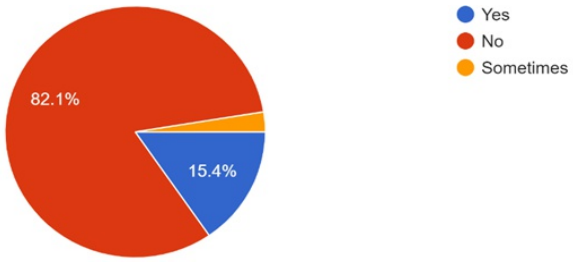


Fig. 9. Respondent response regarding joining queue direction

Fig. 9 shows that 82.1% of patients do not get directions as to where they must join the lines to see the doctors. There are patients who get direction which is shown as 15.4% and 2.5% shows that they do sometimes get the direction.

D. Poor attitude received from the staff respondents

The patients were asked whether they have received any poor attitude from any healthcare workers. The respondent's responses are shown in Fig. 10.

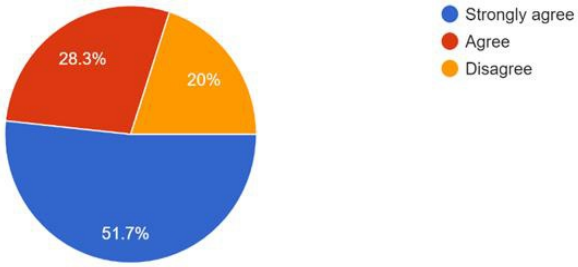


Fig. 10. Poor attitude received from the staff chart presentation

Fig. 10 shows that 51.7% strongly agree, and 28.3% shows that the participants are agreeing and 20% disagree on this question. It shows that the majority strongly agree, and this may scare patients into disclosing their illness due to the attitude of the staff. It is denting the hospital's reputation.

E. Getting the prescribed medication respondents

The question regarding whether the patient received all the medications they needed was also asked. The respondent responses are shown in Fig.11.

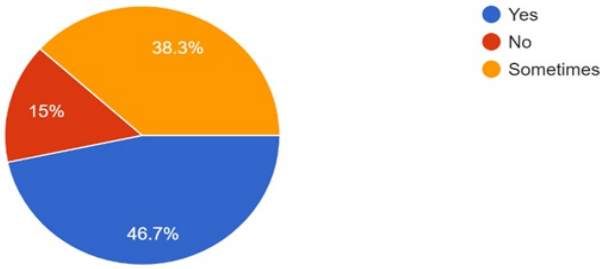


Fig. 11. Getting the prescribed medication.

Fig.11 shows that 46.7% receive the medication and 38.3% do receive the medication sometimes and 15% do not get the meds fully as prescribed, which means they will have to go to the pharmacy and buy and others will not have the medication because they do not have money to purchase over the counter, that is why they went to the public hospital in the first place.

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F. Do clerks attend patients in time respondents

The question of whether clerks attend to patients on time was also asked. The corresponding respondents are shown in Fig. 12.

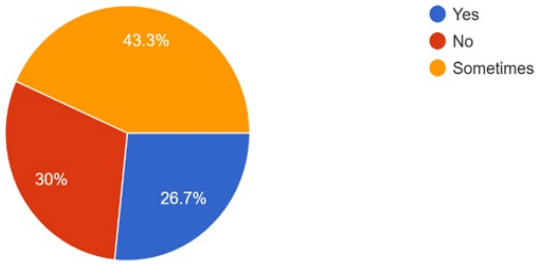


Fig. 12. Respondent responses on whether clerks attend to patients on time.

Fig. 12, shows that 26.7% of clerks attending patients on time, which also is the lowest. This is the reason that makes patients lose hope that they might help. This is contributing to everyone being grumpy because of the long queues. The anecdotal source of lengthy wait times before seeing a doctor is one of the most significant contributors to patient dissatisfaction. It shows that 43.3% attend to patients on time sometimes not often and 30% confirm that they do not attend to patients in time, and this is what makes people not go to the hospital because they will not be assisted in time.

G. Influence of clerks in long queues respondents

Another question was asked by the patient to know whether hospital clerks have an influence on long queues issues in the hospital. The corresponding respondent responses is shown in Fig. 13.

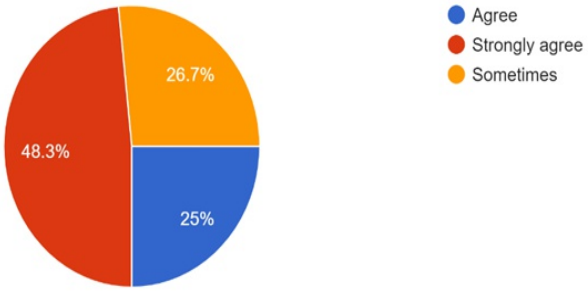


Fig. 13. Influence of clerks in long queues charts presentation

Fig.13 shows that indeed clerks have an influence on the long queues as a result of 48.3% and 25% of patients agreeing on the influence the clerks has and this impact differently on patients because others end up not getting help. The other 26.7% shows sometimes they do have an effect on the queue congestion at the OPD.

H. Staff taking longer to see patients' respondents

The respondents were asked whether they think doctors and nurses take longer to see another patient while they are not doing anything. The response result is shown in Fig.14.

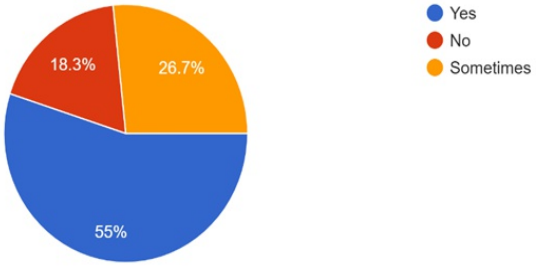


Fig. 14. Patient response on hospital staff time required to see patients.

The staff precisely nurses and doctors take longer to see another patient after one has gone out or after being helped. It shows 55% said yes to doctors and nurses taking long to see patients, 26.7% it shows sometimes, and 18.3% they do not take long to see other patients.

SECTION C: ASSESSMENT OF HEALTHCARE PERSONNEL (COMPULSORY FOR STAFF)

A. Do you frequently work with patients respondents

The staff were asked to know how frequently they work and attend to patients. The corresponding responses are shown in Fig. 15.

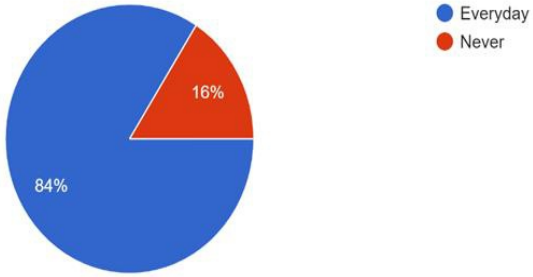


Fig. 15. Respondent responses on how frequently they work with patients.

Fig. 15 shows 84% that the staff frequently work with the patients and 16% show that they do not frequently work them.

B. How do you handle difficult patients respondents

The staff were asked to know how they handle difficult patients. The corresponding responses are shown in Fig. 16.

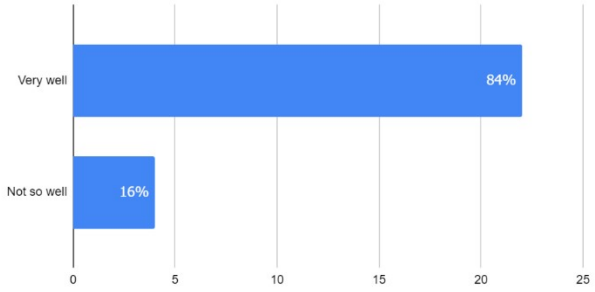


Fig. 16. Respondent responses on how they handle difficult patients

In this figure where the staff handles difficult patients very well with the highest percentage of 84% which is a very good thing and the 16% shows that the staff struggles in handling difficult patients where that on its own waste time.

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C. *How do you work under pressure respondents*

The respondents were asked how well they work under pressure, and the result in Fig.17 shows the respondent's responses.

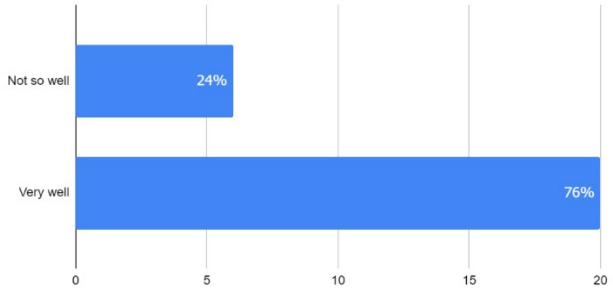


Fig. 17. Respondent responses on how they work under pressure.

This chart has proven that the majority of staff work well under pressure and it has the highest percentage of 76% which also shows that the staff appreciates their work and they are dedicated. With 24% it shows that they do work so well under pressure. In addition, a lack of staff means that the staff that is there is overworked and must work under a variety of intense pressures.

D. *How long do you take to see patients' respondents?*

The questionnaire asked the doctors how long they take to see a patient. The response of the doctors is shown in Fig. 18.

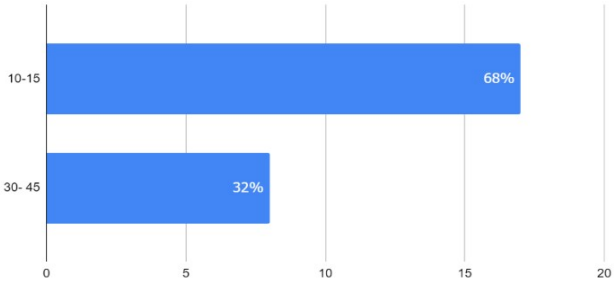


Fig. 18. Respondent responses on how long they take to see a patient

Fig.18 shows that the majority of the staff make sure that they are quick in doing their work so that they can assist a lot of patients as soon as they can as we can see that most staff take up to 15 minutes to help each patient (68%) and other staff take up to 45 minutes (32%). Some experts believe that the consultation time should not be less than 10 minutes.

SECTION D: THE STATISTICS OF FILES STORED ELECTRONICALLY AND A MACHINE SLIPS (NUMBER OF PEOPLE WHO ARE IN SUPPORT)

A. *Do you agree that Machine slip can reduce long queues respondents*

The respondents were asked whether they agreed that utilizing machine slips can reduce long queues. Fig.19 illustrates their responses.

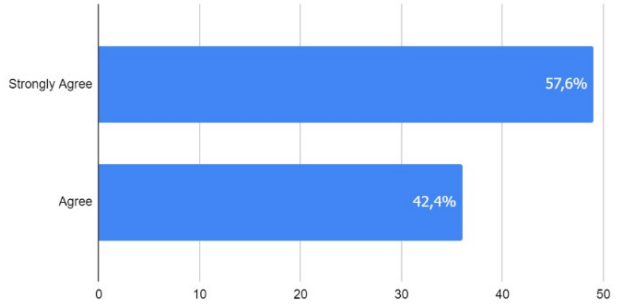


Fig. 19. Respondent responses on the use of Machine slip to reduce long queues.

Fig.19 shows that 57% of people strongly agree on this machine slip can reduce the long queue. And 42,4% show that they agree with this machine. This paper focuses on making the hospital registration process more efficient in order to address the lengthy registration wait time. In this study, the service process, queue strategy, and queue parameters, as well as the registration-related factors of a real hospital, were systematically investigated.

B. *Do you think this machine will help everyone respondents*

Furthermore, the respondents were asked to show their views on whether machine slips can help everyone to reduce their long waiting in the queue.

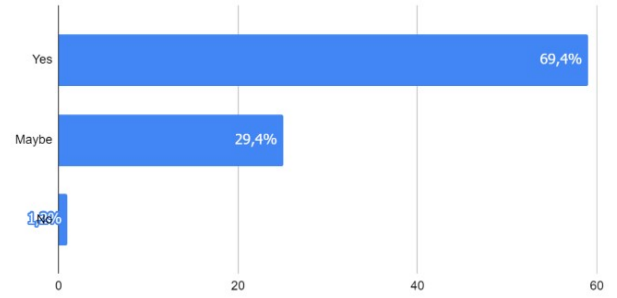


Fig. 20. Respondent responses on the effect of machine slip on the waiting queue

Fig. 20 shows that 69,4% of people think that this machine will help everyone in the hospital in reducing the queues and there is a level of uncertainty as the percentage is shown 29,4% and 3% person do not think that. The majority believe that the machine will assist a lot.

C. *Do you think electronic files can save time and lives respondents*

The question was asked by the respondents at the clinic to know whether electronic files can save time and the lives of patients who needed quick medical attention. The result of the respondents' responses is shown in Fig. 21.

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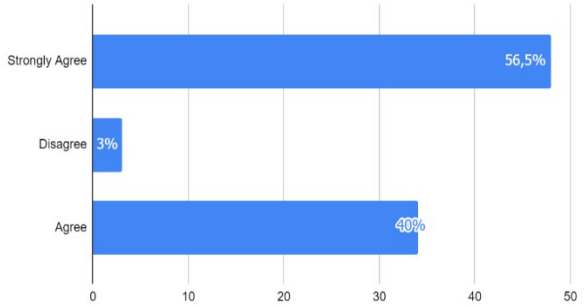


Fig. 21. Respondent responses on the benefit of electronic files on patient time and lives

Fig. 21 shows that electronic files can save time and lives with 56,5% which is also the highest level of which shows that people have a positive belief on files being saved electronically and 40% agree on this idea and 3% people do not believe that it can work.

D. Do you think files being stored electronically is safe respondents

Patients were also asked whether they agree or not that saving their information electronically is safe and secure. The corresponding responses are shown in Fig. 22.

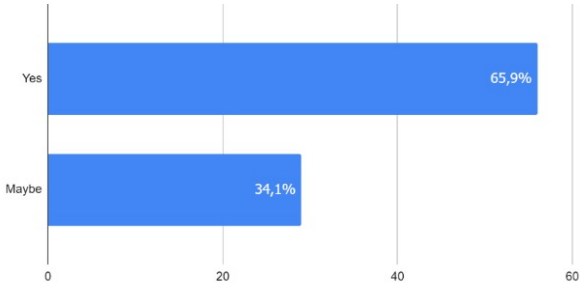


Fig. 22. Respondent responses on file storage.

Fig. 22 shows that the majority do think that files being stored electronically is safe (65,9%). The other 34,1% are so sure about it. Information flow was through many hands, patients' files which can easily get lost or destroyed.

V. Conclusion and Future Work

In conclusion, the effective approach outlined for reducing patient long queues in hospital emergency departments presents a multifaceted strategy aimed at addressing the complex challenges associated with overcrowding and inefficient patient flow. By optimizing triage processes, enhancing resource allocation, streamlining patient flow, improving communication and coordination among healthcare teams, and engaging in community partnerships, hospitals can significantly alleviate patient queues and improve overall ED operations.

This approach emphasizes the importance of adopting a queuing theory to ED management, rather than relying solely on reactive measures to address overcrowding. By implementing these strategies, hospitals can enhance patient outcomes, increase satisfaction levels among patients and healthcare providers, and optimize resource utilization. Moreover, by addressing underlying factors contributing to ED overcrowding, such as access to primary care and preventive care initiatives, hospitals can promote healthier communities and reduce the reliance on emergency services for non-urgent healthcare needs.

Overall, the effective approach to reducing patient long queues in hospital emergency departments underscores the importance of collaboration, innovation, and proactive management in optimizing ED operations and improving the quality of care provided to patients. By embracing these principles and implementing the proposed strategies, hospitals can create a more efficient and patient-centered emergency care environment that meets the needs of both patients and healthcare providers.

Future research could explore the integration of advanced technologies, such as artificial intelligence and machine learning, could enhance predictive analytics models for more accurate forecasting of patient influx and resource allocation. Additionally, collaboration and partnerships with community stakeholders, including primary care providers, mental health facilities, and social services, are critical to addressing underlying factors contributing to ED overcrowding.

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